And the story of the past that you might tell in this situation could easily involve a recognition that you'd actually decided long ago *not* to sell—that no matter how sensible it might be to sell the painting, deep down you've always known that the sentimental connection was just too strong to let it go. The actual past, of course, did not change one bit. Yet a different experience now would lead you to describe a different history.

In the psychological arena, rewriting or reinterpreting the past is commonplace; our story of the past is often informed by our experiences in the present. But in the arena of physics—an arena we normally consider to be objective and set in stone—a future contingency of history makes one's head spin. To make the spinning even more severe, Wheeler imagines a cosmic version of the delayed choice experiment in which the light source is not a laboratory laser but, instead, a powerful quasar in deep space. The beam splitter is not a laboratory variety, either, but is an intervening galaxy whose gravitational pull can act like a lens that focuses passing photons and directs them toward earth, as in Figure 7.3. Although no one has as yet carried out this experiment, in principle, if enough photons from the quasar are collected, they should fill out an interference pattern on a long-exposure photographic plate, just as in the laboratory beam-splitter experiment. But if we were to put another photon detector right near the end of one route or the other, it would provide which-path information for the photons, thereby destroying the interference pattern.

What's striking about this version is that, from our perspective, the photons could have been traveling for many billions of years. Their decision to go one way around the galaxy, like a particle, or both ways, like a wave, would seem to have been made long before the detector, any of us, or even the earth existed. Yet, billions of years later, the detector was built, installed along one of the paths the photons take to reach earth, and switched on. And these recent acts somehow ensure that the photons under consideration act like particles. They act as though they have been traveling along precisely one path or the other on their long journey to earth. But if, after a few minutes, we turn off the detector, the photons that subsequently reach the photographic plate start to build up an interference pattern, indicating that for billions of years they have been traveling in tandem with their ghostly partners, taking opposite paths around the galaxy.